

Foreword

It is certain that, over the next few years, data traffic will dwarf voice traffic on telecommunications networks. Growth in data traffic volumes far exceeds that for voice, and is driven by increased use of applications such as e-mail attachments, remote printing and fileserver access, and, of course, the now widespread World Wide Web.

Advances in the data networking required to support such traffic have been enormous. The pace of development from early X.25 and modem technology through to some of the advanced equipment functionality now available is breathtaking — it is sometimes hard to believe that the practical router is barely ten years old!

The take-up and acceptance of data networking in that time has been astounding. The number of secondary schools in the UK that have local area networks (LANs) in place to interconnect PCs, their printers and, more recently, the Internet, is growing rapidly. Dental practices run networks between the reception and the surgery for handling patient records. The BBC News now includes a WWW site address in the closing titles!

To ensure the continued success of data networking we need ever enhanced public network services and industrial strength network design — both for core network platforms and for end-to-end customer solutions. We need enhancements to the workhorse of the modern customer data network — the router — to support the increased availability and quality of service requirements of networks. We also need accurate synchronisation and ultra-reliable transmission networks upon which we can build. Advantages both to our customers and to the network operator can be gained from asynchronous transfer mode (ATM) through the development of multiservice platforms. Developments in the LAN also continue apace where Ethernet and IP now dominate. Many of these are discussed in depth in this special issue.

In collecting together the material for this edition, we have carefully selected topics we hope to be of wide interest to those in the field as well as to those newer to the discipline and wishing to find out more. Many of the authors are known world-wide for their technical expertise and I am delighted to be able to share with the reader their insight into the field.

I would like to thank all the authors for their submissions, and particularly Debbie Lewis, for her drive and assistance as deputy Guest Editor.

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